THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 49	$7 \times 7 = 49$
2	**(d)** 2^7	Add the exponents: $3 + 4 = 7$
3	**(c)** 17	Add the values of the symbols, subtracting a smaller symbol placed before a larger one
4	**(c)** 90°	The angles of a quadrilateral add up to 360°
5	**(b)** 2346	digit sum 15 is a multiple of 3
6	**(c)** 1287	$99 \times 13 = (100 - 1) \times 13$
7	**(c)** 2 : 3	Divide both terms by 6
8	**(d)** 25%	Multiply by 100: $\frac{1}{4} \times 100 = 25$
9	**(a)** 5	$3^2 + 4^2 = 9 + 16 = 25$, hypotenuse $= 5$
10	**(a)** 2:3	Divide both terms by HCF 6
11	**(d)** 3	Each stage triples the count: $3^1 = 3$
12	**(c)** 21	Sum $= 105$; $\div 5$
13	**(a)** $2x + 5$	2 times x is $2x$; add 5
14	**(d)** 84 cm²	12×7
15	**(c)** 144	$12 \times 12 = 144$
16	**(c)** 2^5	Subtract the exponents: $9 - 4 = 5$
17	**(d)** VIII	8 is written as VIII
18	**(b)** 115°	Adjacent angles of a parallelogram add up to 180°
19	**(b)** 4527	digit sum 18 is a multiple of 9
20	**(a)** 91	$7 \times 10 + 7 \times 3$
21	**(c)** 12	Multiply both terms by 4
22	**(a)** $\frac{1}{5}$	$20\% = \frac{20}{100} = \frac{1}{5}$
23	**(c)** 4	$5^2 - 3^2 = 16$, other leg $= 4$
24	**(d)** 12	Multiplier 4, so $x = 3 \times 4$
25	**(a)** 5	$n + 2 = 5$ (two bases and n rectangles)
26	**(a)** 28	Arrange in order; take the middle value
27	**(c)** 19	$3 \times 5 + 4$
28	**(d)** 30 cm²	$\frac{1}{2} \times 10 \times 6$
29	**(c)** 13	$13 \times 13 = 169$
30	**(b)** 2^6	Multiply the exponents
31	**(a)** 5	Add the powers of 2 where there is a 1
32	**(d)** 58°	Opposite angles of a parallelogram are equal
33	**(a)** 3846	last digit 6 is even
34	**(d)** $3x + 12$	Multiply each term inside by the outside number
35	**(a)** 2 : 3 and 8 : 12	Product of extremes $= 2 \times 12$ equals product of means $= 3 \times 8$
36	**(a)** 25	10/100 × 250
37	**(b)** 3, 4, 5	$3^2 + 4^2 = 25 = 5^2$
38	**(b)** 250	One part $= 400 \div 8 = 50$
39	**(c)** 4	n base vertices + 1 apex $= 4$
40	**(b)** 18	Most frequent value
41	**(c)** $6x + 8$	Add like terms

42	**(a)** 36 cm²	base × height $= 9 \times 4$
43	**(a)** 64	$4 \times 4 \times 4 = 64$
44	**(b)** $\frac{1}{16}$	$a^{-n} = \frac{1}{a^n} = \frac{1}{16}$
45	**(b)** 101	Divide repeatedly by 2 and read the remainders upwards
46	**(b)** 24 cm	$4 \times 6 = 24$
47	**(d)** 21	Add all the digits
48	**(d)** $a^2 + 2ab + b^2$	Standard identity
49	**(b)** 1 : 4	Convert to the same unit first
50	**(c)** 550	Increase = 50; new = 550

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(b)** 12	Hypotenuse $= 5$; perimeter $= 3 + 4 + 5 = 12$
2	**(d)** 35	5×7
3	**(a)** 9	$3n = 9$
4	**(c)** 33	Highest − lowest
5	**(b)** $10x + 2$	Place value: tens × 10 + ones
6	**(a)** 35 cm²	$\frac{1}{2}(8 + 6) \times 5$
7	**(c)** 4	$4^3 = 64$
8	**(c)** 3.2×10^3	One non-zero digit before the decimal point
9	**(b)** 600	Place value = digit x value of its position
10	**(c)** 26 cm	$2(8 + 5)$
11	**(a)** even	Odd + odd = even
12	**(b)** 441	$21^2 = 21 \times 21$
13	**(d)** ₹ 250	One part $= 400 \div 8 = 50$
14	**(b)** 100	SI = PRT/100
15	**(b)** 5	Diagonal$^2 = 9 + 16 = 25$
16	**(c)** 36	10% of $360^\circ = 36^\circ$
17	**(d)** 12	$F + V - E = 2$ gives $E = 6 + 8 - 2 = 12$
18	**(b)** 90	Sum = mean × number of values
19	**(c)** 28	$3 + 4 + 10 + 11$
20	**(d)** 40 cm²	$\frac{1}{2} \times 10 \times 8$
21	**(b)** 25	Sum of first n odd numbers $= n^2 = 25$
22	**(b)** 3400	Move the decimal point right for a positive power and left for a negative power
23	**(a)** 0 to 1	Base 2 uses 2 digits starting from 0
24	**(a)** 10 cm	Diagonals of a rectangle are equal
25	**(b)** 2743	Write the digits in the opposite order
26	**(b)** 17	$3 \times 4 + 5$
27	**(d)** ₹ 8	$48 \div 6$
28	**(d)** 900	SP = MP − discount
29	**(d)** hypotenuse	Side opposite the right angle
30	**(d)** 6	Workers × days is constant: $4 \times 12 = 48$
31	**(d)** 36	A cube has 12 edges: $12 \times 3 = 36$
32	**(c)** one observation	One dot per data value
33	**(a)** 5	Number multiplying the variable
34	**(d)** 25 cm²	Side $= 5$ cm; area $= 5^2$
35	**(d)** 64	$8 \times 8 = 64$
36	**(a)** 3^7	Add the exponents: $2 + 5 = 7$
37	**(a)** L	Learn the seven Roman symbols
38	**(c)** 8 cm	Diagonals of a parallelogram bisect each other
39	**(b)** 4554	A palindrome reads the same from both ends
40	**(d)** 7	The number multiplying x
41	**(b)** ₹ 210	Find the cost of 1 book first
42	**(a)** 60%	Multiply by 100: $\frac{3}{5} \times 100 = 60$

43	**(c)** 10	$6^2 + 8^2 = 36 + 64 = 100$, hypotenuse $= 10$
44	**(a)** 3:5	Divide both terms by HCF 5
45	**(a)** $F + V - E = 2$	Euler's formula
46	**(c)** 13	Sum $= 65$; $\div 5$
47	**(b)** $3x + 4$	3 times x is $3x$; add 4
48	**(d)** 120 cm²	15×8
49	**(d)** 196	$14 \times 14 = 196$
50	**(c)** 3^5	Subtract the exponents: $8 - 3 = 5$

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(b)** 24	Add the values of the symbols, subtracting a smaller symbol placed before a larger one
2	**(a)** 4 sides and 4 vertices	Quad means four
3	**(c)** 12	Smallest common multiple
4	**(b)** 2424	$101 \times 24 = (100 + 1) \times 24$
5	**(a)** 4 : 7	Divide both terms by 5
6	**(b)** $\frac{7}{20}$	$35\% = \frac{35}{100} = \frac{7}{20}$
7	**(a)** 8	$10^2 - 6^2 = 64$, other leg $= 8$
8	**(c)** 30	Multiplier 6, so $x = 5 \times 6$
9	**(c)** 9	Each stage triples the count: $3^2 = 9$
10	**(b)** 24	Arrange in order; take the middle value
11	**(d)** 19	$2 \times 6 + 7$
12	**(c)** 42 cm²	$\frac{1}{2} \times 12 \times 7$
13	**(b)** 17	$17 \times 17 = 289$
14	**(a)** 3^8	Multiply the exponents
15	**(b)** XIX	19 is written as XIX
16	**(d)** 100°	The angles of a quadrilateral add up to 360°
17	**(d)** 8205	digit sum 15 is a multiple of 3
18	**(c)** 324	$6 \times 50 + 6 \times 4$
19	**(d)** 30	Multiply both terms by 6
20	**(d)** 90	20/100 × 450
21	**(a)** 6, 8, 10	$6^2 + 8^2 = 100 = 10^2$
22	**(c)** 200	One part $= 360 \div 9 = 40$
23	**(b)** 7	$n + 2 = 7$ (two bases and n rectangles)
24	**(b)** 7	Most frequent value
25	**(a)** $8x + 8$	Add like terms
26	**(a)** 66 cm²	base × height $= 11 \times 6$
27	**(c)** 125	$5 \times 5 \times 5 = 125$
28	**(d)** $\frac{1}{27}$	$a^{-n} = \frac{1}{a^n} = \frac{1}{27}$
29	**(b)** 6	Add the powers of 2 where there is a 1
30	**(c)** 108°	Adjacent angles of a parallelogram add up to 180°
31	**(b)** 6318	digit sum 18 is a multiple of 9
32	**(d)** $5x + 10$	Multiply each term inside by the outside number
33	**(d)** $4 : 5$ and $12 : 15$	Product of extremes $= 4 \times 15$ equals product of means $= 5 \times 12$
34	**(d)** 1000	Increase = 200; new = 1000
35	**(d)** 24	Hypotenuse $= 10$; perimeter $= 6 + 8 + 10 = 24$
36	**(b)** 60	10×6
37	**(b)** 6	n base vertices + 1 apex $= 6$
38	**(a)** 44	Highest − lowest
39	**(b)** $10x + 3$	Place value: tens × 10 + ones
40	**(b)** 42 cm²	$\frac{1}{2}(10 + 4) \times 6$
41	**(c)** 5	$5^3 = 125$

42	**(c)** 5.8×10^4	One non-zero digit before the decimal point
43	**(b)** 1001	Divide repeatedly by 2 and read the remainders upwards
44	**(c)** 76°	Opposite angles of a parallelogram are equal
45	**(b)** 5712	last two digits 12 are divisible by 4
46	**(a)** $a^2 - 2ab + b^2$	Standard identity
47	**(d)** 1 : 4	Convert to the same unit first
48	**(d)** 600	SI = PRT/100
49	**(a)** 10	Diagonal$^2 = 36 + 64 = 100$
50	**(b)** 54	15% of $360^\circ = 54^\circ$

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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